Research

Two Lines of Inquiry

1. We investigate developmental processes of reading comprehension and content Knowledge.
2. We investigate instructional approaches to support reading comprehension and content knowledge, building on the first line of research.

We study both printed and online/digital texts. Our research often focuses on elementary years and multilingual students.

Current Research Projects

1. Developmental Processes	
1) **Standards of coherence**	Standards of coherence are the implicit and explicit criteria readers use to evaluate whether their understanding of the text is sufficient to meet their reading goals. We develop and validate measures for standards of coherence and examine individual and developmental differences in them (e.g., Hwang et al., 2025 August).
2) **Heterogeneity in literacy and language development**	We investigate heterogeneity in literacy (e.g., word recognition) and language development (e.g., vocabulary) and their contribution to reading comprehension and content learning (e.g., Hwang et al., 2026; Shen & Hwang, 2026). This work includes review studies on digital reading comprehension.

2. Instructional approaches	
1) Lateral reading	Students increasingly interact with online texts to learn. As such, it is critical to support online reading strategies. Lateral reading is an online reading strategy to detect misinformation. We design and test an intervention for lateral reading for

	upper-elementary students (e.g., Son et al., 2026 August).
2) Supporting AI literacy within content-rich literacy instruction	We investigate approaches to supporting AI literacy during content-rich literacy instruction. This work includes Content-rich Literacy and AI literacy via Project-based learning (CLAP).
3) Knowledge revision	Knowledge revision supports knowledge building. We design and examine interventions to support students' knowledge revision during literacy instruction. This includes a project titled Read-Aloud Instruction for SciencE Learning (RAISE Learning; e.g., Hwang et al., 2026).
4) Inference-making	Supporting inference-making is critical as many young students often have difficulties in activating and integrating content knowledge with text. We work with the Inference Galaxy lab to design and test instructional technology in supporting young students' inference-making and content knowledge development.